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by Rahul Sharma

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1.0 Introduction and Project Overview

1.1 Context of the project and scope

Bloomberg's 'European headquarters' is a significant commercial project within the city of London and lies between St. Paul's Cathedral and the Bank of England. Designed by Foster + Partners, it features two interconnected structures covering an area of 3.2 hectares and providing 1.1 million square feet of office area, public space, and retail units. Its purpose was to locate all London employees of Bloomberg in a sustainable place (Dp9.co.uk, 2022).

1.2 Project aim and objectives

The project aimed to construct a consolidated and collaborative European establishment that accommodates the majority of Bloomberg's London workforce, while giving them a working environment through which they can harness an innovative, sustainable, and advanced work culture.

The objectives of the project were as follows.

Firstly, to deliver enhanced quality involving office accommodations and an environment for workforce collaboration.

Secondly, to deploy advanced engineering and building service-based systems that lead to an improvement of a building's operations and carry out better construction prowess.

Thirdly, to integrate public development perspectives for better environmental performance.

1.3 Project timeline and construction phases

It took several years for the project to finish, with the establishment of the Roman Mithraeum in 2010 (Netlify.app, 2021). In the year 2012, Sir Robert McAlpine was selected to complete the construction of this £550 million flagship (Cross, 2012). The construction works comprised two offices, communal spaces, engineering systems, and the entrance to the new Bank Underground station. The offices were opened in October 2017 (Foster Partners, 2017). A depiction of the construction planning can be viewed in Figure 1.

Figure 1: Bloomberg's European Headquarters Construction Design

(Source: <https://www.architectsjournal.co.uk/buildings/building-study-foster-ramps-it-up-at-bloomberg>)

1.4 Significance of the project

The essence of this project stems from the amalgamation of architectural, engineering, and ecological goals with strict urban conditions. This

accomplishment is supported by the score of 98.5% awarded to the project in terms of BREEAM Outstanding certification (Netlify.app, 2021). This provides the basis for analysing change management and risk management in the following sections.

2.0 Change control mechanisms in construction

2.1 Change in construction and other unexpected issues

There may be multiple reasons for variations in projects, including conception and design issues, ineffective management, changes in technology, and external factors (Gunduz and Mohammad, 2020; Padala et al., 2022). Based on an analysis of 3951 changes in 19 construction projects, Padala et al. (2022) identified 85 reasons and 45 consequences of changes in projects. Given this, it is necessary to evaluate the effects of change on cost, schedule, productivity, quality, and stakeholders' relations.

The Bloomberg project is an excellent example of how unforeseen site conditions lead to significant changes. The excavation and restoration of the Roman Temple of Mithras resulted in delaying construction and increasing costs (Foster Partners, 2017). In response to these developments, subcontractors carried out revisions of the initial versions of both temporary and permanent works, which led to the reduction of construction time by more than two months (Robert Bird Group, 2016).

2.2 Change control process involving large-scale projects

The change control process comprises identifying and documenting changes, evaluating cost, schedule, and risk, obtaining approvals, implementing the changes, and evaluating the results (Naji et al., 2022). This is essential because poorly implemented changes can generate the need for rework, cause disruptions, and lead to productivity losses (Love et al., 2019). For instance, 95 construction projects show that the average change orders were equal to 5.9%,

cost overruns were equal to 3.0%, whereas for schedule overruns, the average was 7.4%, demonstrating the usefulness of an efficient change control process (Shrestha et al., 2019).

2.3 Application to the Bloomberg Building

At the initial stage of project execution, changes need to be logged and given a reference number. The project manager is responsible for the programme and technical aspects; the design team evaluates feasibility, and contractors provide information about resources and construction potential. The commercial manager checks that costs and contractual aspects are managed before getting a go-ahead from the client. The importance of collaboration in the process of change orders is derived from the fact that changes have a significant influence on project performance and stakeholders (Nguyen and Do, 2021).

2.4 Control change register

Possible Change

Cause of the change

Expense-related Impact

Programme Impact

Risk Impact

Proposed Responses

Approval by Authorised Personnel

Archaeological activity impacting sequence of construction work

Unforeseen archaeological excavationary circumstances around the construction site

High

High

High

Reprogramming the works already affected; seeking other acceleration options.

Client with Project Manager

Temporary works to be redesigned

Structural activity-related constructions to be accelerated

Medium

Positive depending on success

Medium

Reviewing technical redesign options and overall constructability

Project Manager and Design Team

Method of construction undergoing modification

Sequencing issues or site-related ordeals like hard gravel or porous soil

Medium

Medium

Medium

Looking into alternative methods before final authorisation

Contractor and Design Team

Substitution of construction material or resources

Availability issue involving raw materials and construction materials

Medium

Medium

Medium

Assessing costs, performance, and other factors for equivalent or alternative options

Commercial Manager and Client

Scope adjustment from client-driven agenda

Revision of operational requirements

High

Medium

High

Formal variation assessment and approval

Commercial Manager and Client

Table 1: Control Change Register

(Source: Gunduz and Mohammad, 2020; Padala et al., 2022; Love et al., 2019)

2.5 Commercial implications involving change

Change management is important for safeguarding business interests as well as governance. Each accepted change must be logged together with its reason, scope, cost, schedule influence, potential risk, and approver. The need for such detailed record-keeping is explained by the relationship between change orders and rework and the lower profit margins of contractors. Thus, rework accounted for 0.39% of the project value in the sample of 98 projects (Love et al., 2019). Knowledge of change management allows estimating and accepting changes appropriately, thus controlling costs.

3.0 Risk management and role of a risk register

3.1 Risk management within construction

The inherent uncertainty of construction projects can be attributed to their technical complexity, having multiple parties involved, as well as the changing conditions on-site. Thus, in contrast to a one-time assessment, a continuing process of risk identification, analysis, response and monitoring is necessary (Kallow et al., 2022). Recent empirical research has established a connection between risk identification, monitoring, and mitigation and project success, and has also studied the relationship between risk transparency and coping

ability, including the effectiveness of organisations in coping with uncertainty (Kallow et al., 2022).

3.2 Identifying risk within a Bloomberg-type project

The Bloomberg initiative illustrates the importance of systematic recognition. Archaeological challenges surrounding the construction site caused delays in the programme implementation and a steep rise in costs, forcing contractors to introduce measures to speed up the process. According to the Robert Bird Group, as indicated earlier, the redesigned temporary and permanent works shortened the construction schedule of the affected operation by more than 2 months (Robert Bird Group, 2016). Consequently, the risks that a Bloomberg-type project may face include failure to harmonise the design, uncertainty related to archaeological sites, disruption in supply, escalation of cost, delay in programme, safety occurrences, and changes in legal and sustainability compliance. Furthermore, digital coordination may be helpful in risk management, since studies show that BIM can significantly contribute to risk identification and management even though its use comes together with certain implementation risks (Rehman et al., 2020).

3.3 Assessment of likelihood and consequences

The project can look into a 5*5 qualitative risk matrix, whereby both likelihood and consequence are scored from 1 (being lowest) to 5 (being highest).

Therefore, the risk score can be calculated as follows (Lapidus et al., 2022).

Risk Score=Likelihood*Consequence

Hence, the resulting score requires management attention, serving as a form of authorial assessment for the case, instead of purely being derived from Bloomberg as a form of statistics (Lapidus et al., 2022).

3.4 Risk Assessment Matrix

Likelihood (on the right)

1

2

3

4

5

Consequence (below)

5 (Severe Category)

5

10

15

20

25

4 (Major Category)

4

8

12

16

20

3 (Moderate Category)

3

6

9

12

15

2 (Minor Category)

2

4

6

8

10

1 (Insignificant Category)

1

2

3

4

5

Figure 2: Risk Assessment Matrix

(Source: Self-created)

3.5 Risk Register

Risks

Likelihood (Out of 5)

Consequence (Out of 5)

Risk Score

(L*C)

Residual Risk Likelihood
(Out of 5)

Residual Risk Consequence
(Out of 5)

Residual Risk Score (L*C)

Mitigation

Archaeological site-related uncertainties

4

4

16

2

4

8

Contingency planning post early investigation using specialists for digging and excavation

Design-related coordination failure

3

4

12

2

3

6

BIM-based coordination with multidisciplinary design-based reviewing

Disruption of supply

3

4

12

2

4

8

Alternative supplier choice consideration or earlier procurement of resources

Delay in construction programmes

3

5

15

2

5

10

Monitoring critical path and looking into acceleration-based planning

Expense-related escalation

4

4

16

2

4

8

Forecasting contingency and formal change-related control plans

Table 2: Risk Register

(Source: Self-created)

3.6 Residual Risks

The process of mitigation does not remove uncertainty, but weakens the degree of exposure involving the project. Such residual risk calls for its continuous

ownership and follow-up. This seems important for Bloomberg-type projects because one cannot fully manage archaeology, design interface, and external supply (Nabawy et al., 2020). The risk register should therefore remain a live management tool, since any indicator could trigger a reassessment. This is true as it corresponds to the general recommendations in the literature, which argue for a systematic approach to risk identification instead of using previous experience (Nabawy et al., 2020).

4.0 Lessons from wider construction industry

4.1 Construction industry lessons and response to change

Due to its scale and complexity, Crossrail serves as an appropriate benchmark for the UK construction industry, due to the reliance on strong coordination between companies as well as exchange of information, contractor involvement, and supply-chain participants. To support the BIM strategy, Crossrail made use of a Common Data Environment (CDE), promoting it as a 'single source of truth', linking 3D models with asset management information and scheduling (Taylor, 2018). A better understanding can be gained by viewing Figure 3. This shows that digital technology would be effective only when integrated into governance processes and information management systems, rather than implemented as an independent software solution.

Figure 3: Crossrail's 3D model merging with asset management information and scheduling

(Source: <https://www.linkedin.com/pulse/bim-lessons-australia-from-crossrail-brian-middleton>)

4.2 Technology and change management

The implementation of Building Information Modelling, or BIM, is expected to enhance performance around conflicting design management across

multidisciplinary teams, enabling companies to identify design conflicts without the need for appearing on site and therefore reducing the amount of rework. Chahrour et al. (2020) came up with a cost-benefit analysis of the BIM-driven clash detection process as used on a major building project, demonstrating approximately 20% cost savings resulting from using BIM. Although the finding should not be treated as universally valid, it indicates the significant impact of digital collaboration in a financially measurable aspect. Digital twins allow remote monitoring and prediction; nevertheless, a review of the literature has revealed that the application of the digital twin concept in most case studies examined is actually problematic and limited (Opoku et al., 2021).

4.3 Wider implications related to economic costs

The process of adopting technology entails costs of software, hardware, training, information standards, and organisational change. According to the experience of Crossrail, technology must be put in place, along with processes and infrastructure, while the Information Academy solved knowledge gaps for project teams, contracts, and suppliers (Ahmad 2016). In 2023, the UK Infrastructure and Projects Authority revealed that construction materials were over 40% more expensive compared to January 2020, which caused pressure on budget expenditures and supply chains (Gov.UK, 2024, pp. 9). Thus, commercial managers need to balance implementation costs against the savings from eliminating rework, problems, and ineffective purchases.

4.4 Skills, collaboration, and adaptability

The Information Academy of Crossrail demonstrates that a successful digital transformation relies on people in addition to technologies. It mainly concentrates on People, Process and Technology, and aims to align contractors and suppliers with standard information practices (Ahmad, 2016). Hence, key

competencies for Bloomberg-type projects must include skills such as digital skills, communication, collaboration capabilities, problem-solving and negotiation skills, and even an ability to make commercial judgments. To be successful, the teams must possess relevant technical expertise to transform the information into an integrated decision-making process (Aasen and Klakegg, 2023). While technology can enhance change control, the collaboration of leadership and stakeholders is still important for assessing the impact of the change, finding a solution and implementing changes without transferring uncontrolled costs, time, and quality risks.

5.0 Conclusion

The archaeological considerations impacting Bloomberg's European Headquarters show how unexpected events can interfere with scheduled construction activities and have implications for costs and resources. For change control to be effective, good practices must involve notifying the changes that have taken place, assessing and approving them by project managers, designers, contractors, clients, and commercial managers for assessing their technical, financial and contractual implications. The establishment of change registers is an effective instrument for ensuring accountability and reducing risks arising from unregulated exposure of a project.

The complementary nature of risk management with that of change control lies in the systematic identification and prioritisation of uncertainties involving a construction project. As shown through the 5*5 matrix instance, it is possible to set the level of risk priorities based on the level of probability and consequences; Risk registers allow organising effective mitigation and monitoring actions. It is critical to realise that mitigation does not guarantee the complete elimination of uncertainties. For instance, the archaeological

conditions or disruptions of supply may continue having an impact in the future for the Bloomberg building. Therefore, risk registers must be dynamic instruments that should be used to reevaluate and adjust actions if needed.

The lessons learned from Crossrail have showcased the fact that successful adaptation relies on blending technology with processes and people. BIM and a Common Data Environment can improve information coordination and help with quicker identification of design issues, as training and collaboration can enable people to convert digital data into successful actions. However, the adoption of technology brings with it costs of implementation, the need for skills development, and changes in the organisation. Economic turbulence puts an extra weight on the importance of making informed business decisions.

About the future, digitalisation, AI, BIM, and digital-twin technology create great opportunities for increasing the efficiency of predictive risk management, collaboration and productivity; however, construction companies must deal with cyber security, interoperability, lack of skills, regulatory changes, environmental demands, and uncertainty in the supply chain. Thus, construction change management involves ensuring proper governance, active risk management, using the best technologies, good teamwork, and necessary practical experience for preparing the companies for both possible and unexpected changes while keeping in mind the costs, deadlines, quality, and profitability of the projects.

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